Skills Summary

Transformations with Multiple Parameters

Use this reference while completing your worksheet or when studying.

The parameter map: $g(x) = a\,f\big(b(x-h)\big) + k$

a	**outside**	vertical stretch by $\lvert a\rvert$; reflect over the x-axis if $a < 0$	does what it says
k	**outside**	shift up k	does what it says
b	**inside**	horizontal stretch by $1/b$	does the opposite
h	**inside**	shift right h	does the opposite

Factor first. h can only be read once the inside is written as $b(x-h)$: $f(2x-6) = f\big(2(x-3)\big)$ shifts 3, not 6.

Skill 1 — The input chain (table-transform)

Order of operations on a single input: $x \to (x-h) \to b(x-h) \to f(\ldots) \to a\cdot f(\ldots) \to a\cdot f(\ldots)+k$.

Build a column for each arrow. The parent function runs in the *middle* — everything to its left is an input move, everything to its right is an output move.

Example: $f(x) = x^2$, $g(x) = 3f(x+1) - 2$. Find $g(1)$.
Step 1. Two parameters act, one inside and one outside, so I walk the input through the chain in order rather than guessing at the end.
Step 2. Input move: $1+1=2$. Parent: $f(2)=4$. Output moves: $3(4)=12$, then $12-2=10$. $\Rightarrow$ $\boldsymbol{g(1) = 10}$

Try it: $f(x) = x^2$, $h(x) = -f(x-2) + 6$. Find $h(5)$.

check: $\boldsymbol{h(5) = -3}$

Skill 2 — Factor before you read (parameters-from-equation)

Rule: factor b out of the *entire* inside before naming the shift — $bx + c = b\left(x + \frac{c}{b}\right)$, so the shift is c/b, not c.

For $y = A\sin\big(B(x-C)\big)+D$: amplitude $\lvert A\rvert$, period $2\pi/B$, phase shift C, midline $y = D$.

Example: Rewrite $y = 2\sin\left(3x + \frac{\pi}{2}\right)$ in the form $A\sin\big(B(x-C)\big)$.
Step 1. The shift is unreadable while 3 still multiplies only part of the inside, so the first move is to factor $B = 3$ out of everything.
Step 2. $3x + \frac{\pi}{2} = 3\left(x + \frac{\pi}{6}\right)$, since $3\cdot\frac{\pi}{6} = \frac{\pi}{2}$. So $A = 2$, $B = 3$, $C = -\frac{\pi}{6}$: shifted $\frac{\pi}{6}$ *left*, not $\frac{\pi}{2}$. $\Rightarrow$ $\boldsymbol{2\sin\left(3\left(x + \frac{\pi}{6}\right)\right)}$

Try it: Rewrite $y = 5\cos(4x - \pi)$ in the form $A\cos\big(B(x-C)\big)$.

check: $\boldsymbol{5\cos\left(4\left(x - \frac{\pi}{4}\right)\right)}$

Skill 3 — Composition order (composition-order)

Rule: in $(u \circ v)(x) = u(v(x))$ the *inner* map runs first. Swapping the order changes the function, because a stretch applied after a shift scales that shift, while a stretch applied before it does not.

Example: $s(x) = x + 5$ (shift), $d(x) = 3x$ (stretch). Simplify $(d \circ s)(x)$.
Step 1. The inner map s runs first, so the shift happens and the stretch then acts on the already-shifted value.
Step 2. $d(s(x)) = d(x + 5) = 3(x + 5)$. Distributing shows the shift was scaled: the graph moves 15, not 5. $\Rightarrow$ $\boldsymbol{3x + 15}$ (whereas $(s \circ d)(x) = 3x + 5$)

Try it: $t(x) = x - 2$, $w(x) = 4x$. Simplify $(w \circ t)(x)$.

check: $\boldsymbol{4x - 8}$

(!) Watch out: Reading a shift straight off an unfactored inside is the most common multi-parameter error — always divide the constant by b (or factor, which does it for you). And when you undo parameters to solve an equation, peel them off from the *outside in*: k first, then a, then the parent, then b and h.